

REPLACE YOUR PHONE WITH AN **AI-POWERED** SMARTPHONE

Ayushee Sharma

Adancements in technology over the last decade have led to phones getting smarter with time. While artificial intelligence (AI) already powers several features on smartphones like maps and virtual assistants, the number of smartphones with dedicated AI-powered chips is growing at a fast pace.

With incorporation of AI, phones can now take their own decisions based on surroundings without having to be programmed to do the same. When computations are done locally, it ensures lower latency with power optimisation and better security as data is not uploaded to the cloud over the Internet. Other benefits include personalisation and an enhanced augmented reality (AR) experience.

Making the right choice

With so many smartphones out there, it becomes confusing to choose the right one until you know the parameters to consider beforehand. Some of these parameters are discussed in the next section.

Processor. Processors are responsible for on-device AI capabilities. Manufacturers are specifically focused on processors for AI-based imaging, connectivity, security and mul-

titasking. This enables improved virtual assistants, secure fingerprint scanners and face IDs for unlocking. Many of the functions can be performed even without an Internet connection.

Processor speed, which denotes the maximum speed of operation, should be high for better performance. A neural processing unit (NPU) uses machine learning to learn over time and automate different tasks. A neural engine ensures fast performance and enhanced efficiency.

Display. Higher resolution, such as that in quad high-definition (QHD), when compared to HD, means better display quality. Screen-to-body ratio percentage denotes the amount of surface the screen represents as compared to the entire smartphone. A higher percentage means a larger screen.

Operating system. An operating system (OS) is the main software on which the phone operates. Google's Android OS is generally found on most devices, although iPhones use iOS. Huawei officially unveiled its Harmony OS in 2019 to replace Android OS on its devices.

Battery. With AI-powered phones, the ability to perform battery-intensive tasks is higher. It is necessary to choose a higher mAh battery to prevent the phone from draining easily. Many phones come with options such as Qi wireless charging and reverse charging that shares power with another device. Some devices also limit battery usage for applications that are not used often.

Connectivity. While dual-band (2.4GHz + 5GHz) Wi-Fi connectivity is common in most phones, devices with next-generation standard, Wi-Fi 6, are also coming up in the market. When it comes to Bluetooth, latest phones support version 5.0, which has better speed

and higher bandwidth as compared to previous versions.

Storage. Working with a processor, good RAM is essential for higher speed and seamless operation of tasks. Along with it, high internal storage (ROM) ensures enough space for different applications. Many phones come with the option to expand internal storage, while variants with different sizes are also available in the market.

Camera. This is one of the most prominent features where AI is used for such tasks as better facial recognition and object detection. These devices come with multiple rear cameras along with a high-resolution front camera. Smart HDR gives more highlight and shadow detail to images.

Telephoto camera is good for portrait and distant photography. For selfies, there is portrait mode with depth control option. AI algorithm adjusts camera settings for better selfies. Machine learning is used for background blur. All these give pictures enhanced clarity and ensure no pixelation.

Market options

Here are some of the options that you may like to consider while making the purchase. Prices may vary on different e-commerce websites.

Honor View 20. Honor View 20 (15.7cm x 0.8cm x 7.5cm; 181gm) has a 16.25cm FHD display with 91.8 per cent screen-to-body ratio. Powered by 2.6GHz Kirin 980 AI octa-core processor, it has Android v9.0 (Pie) OS with Magic UI 2.0 on top.

